



GSA ANNUAL SCIENTIFIC MEETING ONLINE
WORKSHOP SERIES 2026

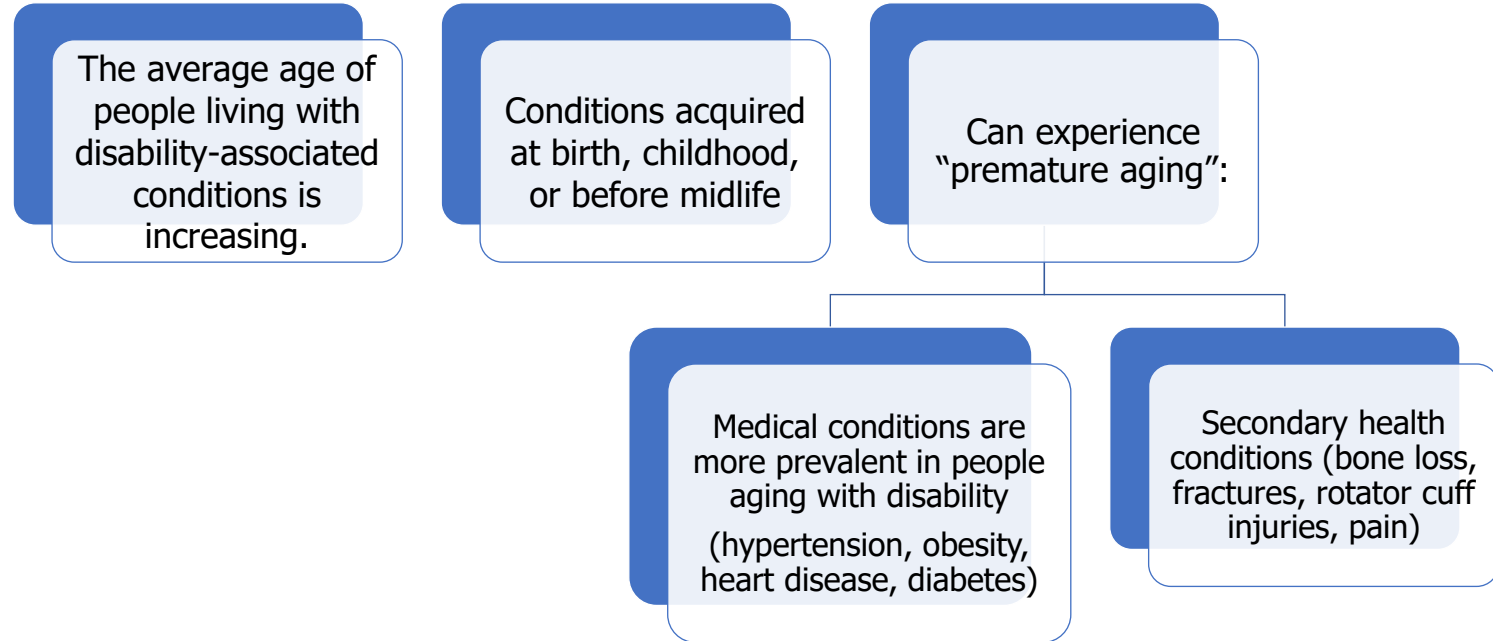
Maintaining Health while Aging with Disability:

The Role of the Neighborhood Environment

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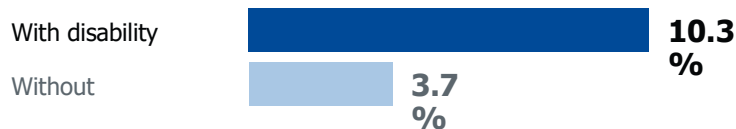
Aging with disability



Health disparities and disability

Health conditions are more common in adults with disability:

Cardiovascular disease prevalence is almost three times higher



Diabetes prevalence is twice as high



Health behaviors and disability

Compared to those without disability, adults with disability are:

2×

Twice as likely to be current smokers

+20%

20% more likely to be sedentary

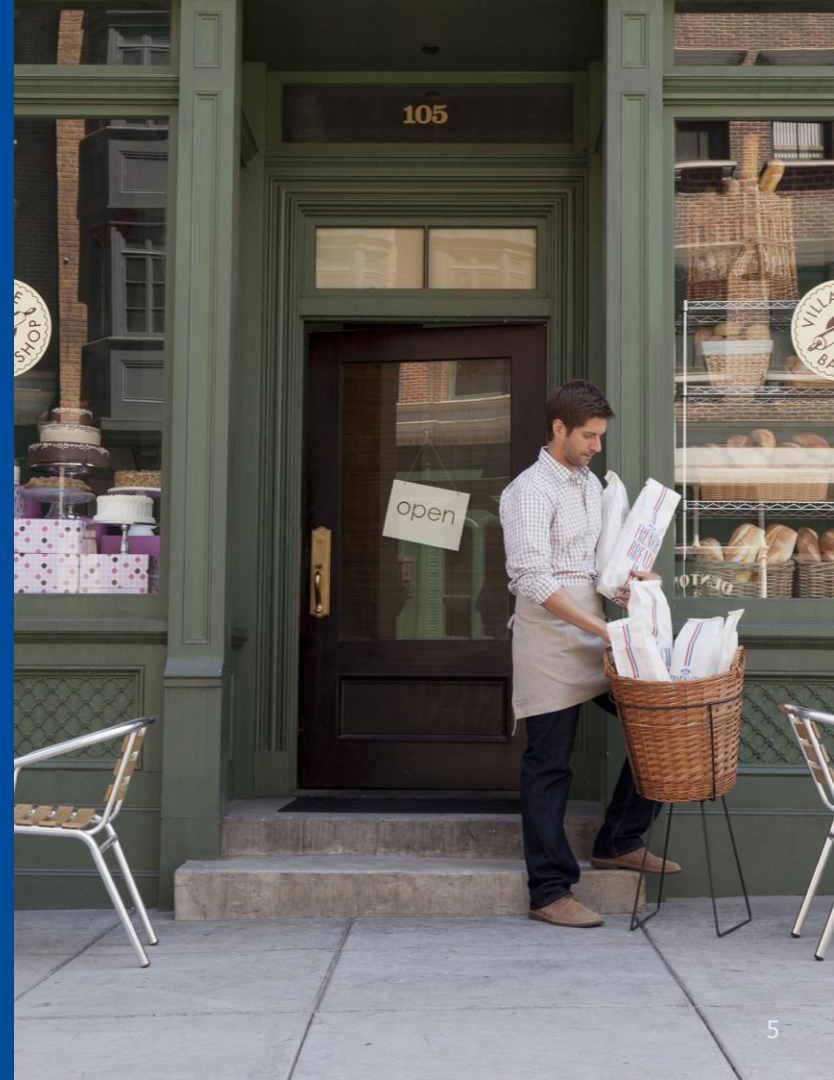
+10%

10% more likely to be obese

Neighborhoods and Incident Cardiometabolic Disease in People Aging with Disability

Clarke et al., 2025. *American Journal of Preventive Medicine*, 69(5).

doi.org/10.1016/j.amepre.2025.108001

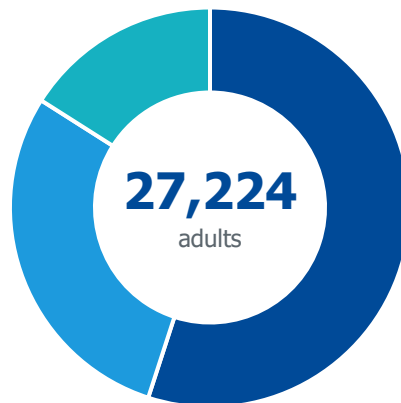


Data source: Medicare claims

20% random sample of de-identified administrative claims for Medicare beneficiaries (2008 to 2016)

Received Social Security Disability Income prior to enrollment in Medicare

Mean age 45 years (± 14)



- 55%**
Cerebral palsy or spina bifida
- 29%**
Spinal cord injury (plegia)
- 16%**
Multiple sclerosis

Measures

OUTCOME

3-year incident cardiometabolic disease (ICD-10-CM codes)

- Cardiovascular conditions (e.g., hypertension, heart failure, cardiac dysrhythmias)
- Type 2 diabetes mellitus
- Chronic kidney disease

EXPOSURE

Neighborhood context

- Residential ZIP code linked to its ZIP Code Tabulation Area (ZCTA)
- Linked to data in the National Neighborhood Data Archive (NaNDA)



NaNDA's neighborhood measures (ZCTA)

DOMAIN	MEASURES
 Recreation environment	Recreational facilities (e.g., gyms, fitness centers, pools), Parks
 Consumables environment	Grocery stores, fast-food restaurants, liquor stores, tobacco stores
 Healthcare resources	Outpatient care (e.g., physicians, physical and occupational therapists)
 Traffic volume	Average daily non-highway traffic counts per intersection

Cox regression models

- 01 Time (days) to incident disease
- 02 Account for clustering within ZCTAs with robust sandwich covariance matrix
- 03 Free of any previous cardiometabolic disease

Adjusting for covariates

Age

Sex

Race/ethnicity

Comorbid conditions (Elixhauser Comorbidity Index, ICD-9 codes)

HMO coverage

Neighborhood population density

Neighborhood SES (Affluence, Disadvantage)

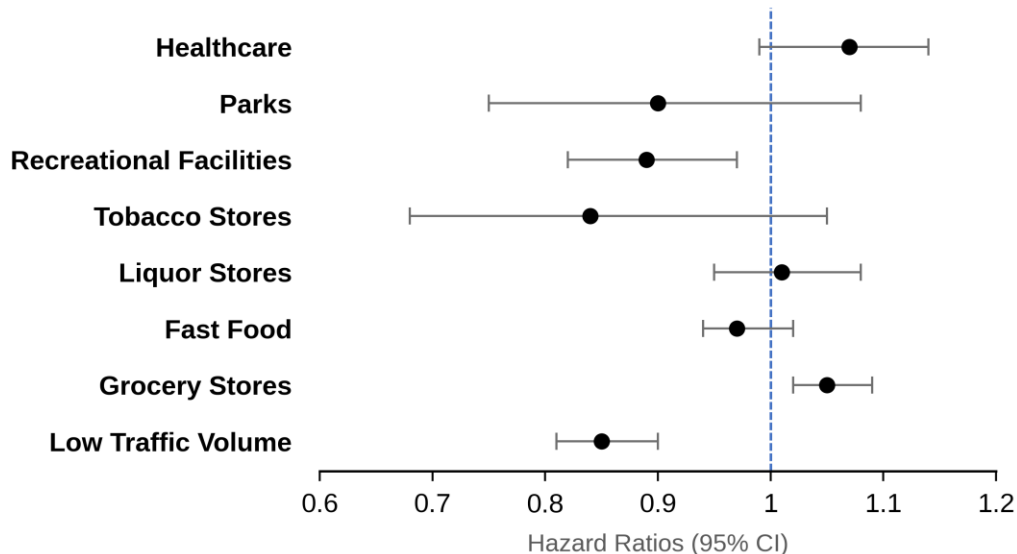
Results: Cardiometabolic incidence (3-year)



Results

Neighborhood resources and 3-year incident cardiometabolic disease

Models adjust for age, sex, race/ethnicity, comorbidity, HMO, neighborhood SES, population density.



Implications

Recreational facilities

Living in a neighborhood with a greater density of recreational facilities is associated with:

11% reduced risk of cardiometabolic disease in adults aging with disability

Critical that recreational facilities are accessible and welcoming to people as they age with disability



Grocery stores

Living in a neighborhood with a greater density of grocery stores is associated with:

5% increased risk of cardiometabolic disease in adults aging with disability

Healthy food in grocery stores is often more expensive;
Barriers in grocery stores (narrow aisles, tall shelves) may deter shopping



Traffic

Living in a neighborhood with lower traffic is associated with:

15% reduced risk of cardiometabolic disease in adults aging with disability

People with disability often rely on public transit or pedestrian walkways, and may not have resources to own/adapt a vehicle

Heavy traffic is the top environmental factor in injuries and deaths among disabled pedestrians (Schwartz, 2022)





**Identifying neighborhood
factors that could mitigate
cardiometabolic risk is critical
for supporting healthy aging
with disability**